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Week 3: Class Activity (21.05) — Report

Student Number: 110195911 **Student:** Lei Jiang **Date:** 2026-05-23

Task 1 — Add phone_number field to the Student model

Source code

classroom/models.py

```
from django.db import models

class Student(models.Model):
    name = models.CharField(max_length=100)
    email = models.EmailField(unique=True)
    phone_number = models.CharField(max_length=20, blank=True, default="")
    join_date = models.DateField(auto_now_add=True)

    def __str__(self):
        return self.name
```

Migration generated

```
$ python manage.py makemigrations classroom
Migrations for 'classroom':
  classroom/migrations/0002_student_phone_number.py
    + Add field phone_number to student

$ python manage.py migrate
Running migrations:
  Applying classroom.0002_student_phone_number... OK
```

Result in the browser

<http://127.0.0.1:8000/students/> — the table now has a **Phone** column.

127.0.0.1:8000/students/

Name	Email	Phone	Joined
Alice Wong	alice@uwindsor.ca	+1-519-111-1111	May 23, 2026
Bob Singh	bob@uwindsor.ca	+1-519-222-2222	May 23, 2026

Task 2 — Create [/students/recent/](#) page (last 7 days)

Source code

[classroom/views.py](#)

```
from datetime import timedelta

from django.shortcuts import render
```

```

from django.utils import timezone

from .models import Student

def recent_students(request):
    days = 7
    cutoff = timezone.now().date() - timedelta(days=days)
    qs = Student.objects.filter(join_date__gte=cutoff).order_by("-
join_date")
    return render(request, "classroom/recent_students.html", {
        "students": qs,
        "days": days,
        "count": qs.count(),
        "is_empty": not qs.exists(),
    })

```

config/urls.py (new route)

```

from classroom.views import student_list, recent_students

urlpatterns += [path("students/recent/", recent_students, name="student-
recent")]

```

classroom/templates/classroom/recent_students.html (excerpt)

```

<h1>Recent Students</h1>
<p class="sub">Joined in the last {{ days }} day{{ days|pluralize }} · {{
count }} result{{ count|pluralize }}</p>
<table>
  <thead><tr><th>Name</th><th>Email</th><th>Phone</th><th>Joined</th></tr>
</thead>
  <tbody>
    {% for s in students %}
      <tr>
        <td>{{ s.name }}</td>
        <td>{{ s.email }}</td>
        <td>{{ s.phone_number|default:"-"}</td>
        <td>{{ s.join_date|date:"M d, Y" }}</td>
      </tr>
    {% empty %}
      <tr><td colspan="4" class="empty">No students joined in the last {{
days }} days.</td></tr>
    {% endfor %}
  </tbody>
</table>

```

Result in the browser

<http://127.0.0.1:8000/students/recent/> — shows only students whose `join_date` is within the last 7 days.

127.0.0.1:8000/students/recent/

Recent Students

Joined in the last 7 days · 2 results

Name	Email	Phone	Joined
Alice Wong	alice@uwindsor.ca	+1-519-111-1111	May 23, 2026
Bob Singh	bob@uwindsor.ca	+1-519-222-2222	May 23, 2026

Note. The recent page lists the same two rows as `/students/` because Query 3 (DELETE) in Task 3 removed the two backdated students (Carla, Dan, joined 2026-05-03). The two remaining students (Alice, Bob) were created today, so they fall inside both queries. This is the expected outcome and demonstrates that the DELETE query actually ran.

Task 3 — Three shell queries (filter, order, delete)

Source code — `shell_queries.py`

```
from datetime import timedelta
from django.utils import timezone
from classroom.models import Student

# Seed data
Student.objects.all().delete()
s1 = Student.objects.create(name="Alice Wong", email="alice@uwindsor.ca",
                             phone_number="+1-519-111-1111")
s2 = Student.objects.create(name="Bob Singh", email="bob@uwindsor.ca",
                             phone_number="+1-519-222-2222")
s3 = Student.objects.create(name="Carla Mendes", email="carla@uwindsor.ca",
                             phone_number="+1-519-333-3333")
s4 = Student.objects.create(name="Dan Patel", email="dan@uwindsor.ca",
                             phone_number="+1-519-444-4444")

# Backdate two students so the recent-7-days filter has something to exclude
```

```

old = timezone.now().date() - timedelta(days=20)
Student.objects.filter(pk__in=[s3.pk, s4.pk]).update(join_date=old)

# Query 1 – FILTER
qs_filter = Student.objects.filter(name__icontains="a")

# Query 2 – ORDER
qs_order = Student.objects.order_by("-join_date")

# Query 3 – DELETE
cutoff = timezone.now().date() - timedelta(days=7)
deleted, info = Student.objects.filter(join_date__lt=cutoff).delete()

```

How to run

```
uv run manage.py shell < shell_queries.py
```

Result (shell output)

```

--- All students after seeding ---
9  Alice Wong    alice@uwindsor.ca    +1-519-111-1111    2026-05-23
10 Bob Singh    bob@uwindsor.ca      +1-519-222-2222    2026-05-23
11 Carla Mendes carla@uwindsor.ca    +1-519-333-3333    2026-05-03
12 Dan Patel    dan@uwindsor.ca      +1-519-444-4444    2026-05-03

--- Query 1 (FILTER): name__icontains='a' ---
9  Alice Wong    alice@uwindsor.ca
11 Carla Mendes carla@uwindsor.ca
12 Dan Patel    dan@uwindsor.ca

--- Query 2 (ORDER): order_by('-join_date') ---
9  Alice Wong    2026-05-23
10 Bob Singh    2026-05-23
11 Carla Mendes 2026-05-03
12 Dan Patel    2026-05-03

--- Query 3 (DELETE): join_date__lt=2026-05-16 ---
Deleted: 2 {'classroom.Student': 2}

--- Remaining students after delete ---
9  Alice Wong    2026-05-23
10 Bob Singh    2026-05-23

```

```
+ week3 git:(main) uv run manage.py shell < shell_queries.py
13 objects imported automatically (use -v 2 for details).
```

```
--- All students after seeding ---
```

```
9 Alice Wong alice@uwindsor.ca +1-519-111-1111 2026-05-23
10 Bob Singh bob@uwindsor.ca +1-519-222-2222 2026-05-23
11 Carla Mendes carla@uwindsor.ca +1-519-333-3333 2026-05-03
12 Dan Patel dan@uwindsor.ca +1-519-444-4444 2026-05-03
```

```
--- Query 1 (FILTER): name__icontains='a' ---
```

```
9 Alice Wong alice@uwindsor.ca
11 Carla Mendes carla@uwindsor.ca
12 Dan Patel dan@uwindsor.ca
```

```
--- Query 2 (ORDER): order_by('-join_date') ---
```

```
9 Alice Wong 2026-05-23
10 Bob Singh 2026-05-23
11 Carla Mendes 2026-05-03
12 Dan Patel 2026-05-03
```

```
--- Query 3 (DELETE): join_date__lt=2026-05-16 ---
```

```
Deleted: 2 {'classroom.Student': 2}
```

```
--- Remaining students after delete ---
```

```
9 Alice Wong 2026-05-23
10 Bob Singh 2026-05-23
```

How to reproduce

```
cd code/week3
uv venv
uv pip install django
uv run manage.py migrate
uv run manage.py shell < shell_queries.py      # seed + run the three
queries
uv run manage.py runserver
# open http://127.0.0.1:8000/students/
# open http://127.0.0.1:8000/students/recent/
```